

Sex differences and gendered bodies

At the centre of commonsense thinking about gender is the idea of natural difference between women and men. A whole industry of pop psychology tells us that women and men are naturally opposites in their thinking, emotions and capacities. The most popular book in this genre, which assures us that men and women are like beings from different planets, has sold *50 million* copies and is translated into 50 languages. Other books in this genre, and endless articles in popular magazines, tell us that men and women communicate in different ways, that boys and girls learn differently, that hormones make men into warriors, or that ‘brain sex’ rules our lives.

Most of the claims in these books are, in scientific terms, complete nonsense – refuted by a mass of research evidence. The US psychologist Janet Hyde (2005), the leading authority on gender difference research, points out that the pop-psychology doctrine of natural difference is harmful to children's education, to women's employment rights, and to all adults' emotional relationships. Clearly we need better ways of thinking about differences and bodies. The development of gender studies now provides some of the necessary tools.

Reproductive difference

Why is there any difference at all between women's and men's bodies? Humans share with many other species, plants as well as animals, the system of sexual reproduction – a method of reproducing which allows genetic information from two individuals to be combined, rather than just one to be copied. Sexual reproduction is itself a product of evolution, perhaps 400 million years old. Life forms existed earlier that reproduced by cell division from one individual; many species still do this, including bacteria and rotifers. Other species, including orchids, ferns and grasses, reproduce both sexually and asexually. Gardeners exploit the reproductive capacities of plants when they take cuttings from a neighbour's garden. Biologists debate why sex evolved, for this odd scheme has some evolutionary disadvantages, such as energy use or unfavourable genetic recombination. One speculation is that sex may have evolved because sexual reproduction allows faster change; another is that it prevents the accumulation of harmful mutations.

Sexual reproduction does not require bodies to be specialized by sex. Among earthworms, for instance, each individual is hermaphrodite, producing both sperm and ova (eggs), and thus every worm is able to perform both male and female functions. In other species, individuals produce either sperm or ova but not both. Their bodies are to some extent ‘dimorphic’, i.e. in a

given species there exist two forms. Humans are among these species.

Genetic information is encoded in DNA and carried on chromosomes, microscopic structures within the nucleus of each cell in a plant or animal. In sexual reproduction, the genetic information that is combined at fertilization comes half from a female, in the egg nucleus, and half from a male, in the sperm nucleus. Human cells have 46 chromosomes, which come in pairs. One pair, the so-called 'sex chromosomes', influences the development of the body's male and female sexual characteristics. Females have two X chromosomes in this pair, males have one X and one Y chromosome. Under the influence of the genetic information here, and given the usual environmental conditions, male and female bodies develop specialized organs – wombs, testes, breasts – and certain differences in physiology, such as the balance of hormones circulating in the blood, and the menstrual cycle in women.

Among mammals, females not only produce ova but also carry foetuses in a protective womb (except for monotremes such as the platypus, which lay eggs). They feed infants with milk from specialized organs – in humans, women's breasts. Among some mammal species, but not all, males have extra bulk, or extra equipment: the antlers of male deer, for instance. Humans are mammals with well-differentiated reproductive systems, but modest physical differences between sexes in other respects. Human males do not have antlers.

In several ways, human bodies are not fully dimorphic. First, there are a considerable number of intersex categories, such as females lacking a second X chromosome, males with an extra X chromosome, anomalous or contradictory hormonal patterns, and a surprising variety of un-standard forms taken by the internal and external genitals. The biologist Anne Fausto-Sterling (2000: 51) estimates that the different intersex groups, taken together, may account for 1.7 per cent of all births; others think the proportion is lower (Sax 2002). There is continuing debate on how broad the category should be, what terminology applies, and how intersex people should be recognized in law (Holmes 2012). A major debate concerns a new term proposed in 2006 by European and US endocrinologists called 'Disorder of Sexual Development'; intersex advocates are worried about the way medical terms such as 'disorder' influence forms of treatment.

Katrina Karkazis's *Fixing Sex: Intersex, Medical Authority, and Lived Experience* (2008) traces the history of medical treatments of people diagnosed intersex. Drawing on interviews in the United States with clinicians, adults with intersex conditions and their parents, she shows that over five decades the health movement has changed medical diagnoses of intersex, but most clinicians still hold heteronormative and homophobic ideas about genital appearance. These views often mean doctors encourage parents to choose surgical procedures for their children. Interviews with patients in Karkazis's book and a collection of life stories published in Catherine Harper's *Intersex* (2007) offer a strong critique of the rush to 'correct' these variations by surgery on small children.

Second, physical differences between male and female change over the lifespan. In the early stages of development, male and female bodies are relatively undifferentiated; there are only small differences between a 2-year old girl and a 2-year-old boy. Even the external reproductive organs – penis, clitoris, scrotum and labia – develop embryonically from a

common starting point. In a number of ways, male and female bodies also become more similar in old age, for instance in their hormonal balance.

Third, even in early adulthood, the physical characteristics of males as a group, and females as a group, overlap extensively. Height is a simple example. Adult males are on average a little taller than adult females, but the diversity of heights within each group is great in relation to the average difference. Therefore a very large number of individual women are taller than many individual men. European cultures tend not to notice this physical fact because of social custom. When a man and a woman form a couple, they usually pick partners who show the 'expected' difference in height.

A more complex example is the brain – the site of a great deal of recent debate on sex differences. There are some differences in brain anatomy and functioning between women and men, for instance in the tendency to use particular areas of the brain in language processing. But the differences are fewer, and less reliably established, than aggressive popular accounts of 'brain sex' suggest. In many areas of brain anatomy and functioning, there are no significant sex differences. Where there are differences, these may be caused by different behaviours rather than causing them. Brain research now places a lot of emphasis on 'brain plasticity', the capacity of the brain to form new neural connections and lose old ones, i.e. to learn and change. As the neuroscientist Lesley Rogers (2000: 34) puts it: 'The brain does not choose neatly to be either a female or a male type. In any aspect of brain function that we can measure there is considerable overlap between females and males.' As we shall see, this is also a key point about human behaviour.

Conflicting accounts of difference

The fact of reproductive difference between male and female humans is hardly controversial, but its significance certainly is. On this question, approaches to gender diverge sharply. Some treat the body as a kind of machine that manufactures gender difference; some treat the body as a kind of canvas on which culture paints images of gender; some try to staple the machine and canvas images together. None of these is a satisfactory way of understanding the problem.

Many writers about gender assume that reproductive difference is directly reflected in a whole range of other differences: bodily strength and speed (men are stronger and faster), physical skills (men have mechanical skills, women are good at fiddly work), sexual desire (men have more powerful urges), recreational interests (men love sport, women gossip), character (men are aggressive, women are nurturant), intellect (men are rational, women have intuition), and so on. It is widely believed that these differences are large, and that they are 'natural'.

The claim that natural difference provides the basis for the social pattern of gender takes many forms. One is that men dominate in society because, with their higher levels of testosterone, they have a hormonal 'aggression advantage' in competition for top jobs. Therefore society needs patriarchy – Steven Goldberg claimed in *Why Men Rule* (1993) – to protect women from failure!

Arguments in 'sociobiology' and 'evolutionary psychology' deduce social gender from reproductive strategies. From this vaguely Darwinian starting point, theorists have deduced human kinship loyalties, mothers' commitment to their children, husbands' sexual infidelity, women's coyness, men's interest in pornography, male bonding and a remarkable range of other gender patterns. In *Male, Female* (1998), David Geary tries to link psychological research on sex differences with Darwin's concept of 'sexual selection' (the choice of mates in sexual reproduction) as a mechanism of evolution. For each topic where a sex difference can be located, Geary offers an account of how it *might* be linked to sexual selection, i.e. how humans choose, win and control mates. In *Sex, Genes and Rock 'n' Roll* (2011), Australian evolutionary biologist Rob Brooks applies the same view to popular culture. He argues that rock 'n' roll music can be understood as a reproductive agenda for men. That is, fame serves the male search for access to sexual partners.

Models of the body as a gender machine are mainly advanced by men, and have often been used to defend the existing gender order and ridicule feminist ideas about gender roles. However, there are also feminist arguments which present bodies as direct sources of gender difference. US feminists in the 1980s often saw male aggression and female peacefulness as natural. The terms 'male violence' and 'male sexuality', which became common at this time, implicitly linked behaviour to the body, and some activists directly identified the penis as the source of male power. This perspective reinforces the view that bodies are a separate aspect of our subjectivity that causes mental and social phenomena.

The idea of natural difference runs into difficulties on several fronts. Sociobiological explanations of human kinship, for instance, foundered when the predictions from genetics failed to match the realities of kinship systems actually documented by anthropologists (Sahlins 1977). The explanation of gender hierarchy by a hormonal 'aggression advantage' fails when it is discovered that higher testosterone levels *follow from* social dominance as much as they precede it (Kemper 1990). The 'evolutionary psychology' arguments are based on an unrealistic individualism, which takes no account of institutionalized gender arrangements. For instance, in discussing the higher levels of violence among men than among women, all that Geary (1998) can see is male vs male competition for reproductive resources. He cannot see armies, governments, insurgencies, mafias or cultural definitions of manhood – let alone football. But the most striking problem is that these arguments, while appealing to science, are actually entirely based on speculation. Not one sex difference in psychological characteristics has actually been *shown* to result from evolutionary mechanisms.

It is clear that bodies are affected by social processes. The way our bodies grow and function is influenced by food distribution, sexual customs, warfare, work, sport, urbanization, education and medicine, to name only the most obvious influences. And *all* these influences are structured by gender. So we cannot think of social gender arrangements as just flowing from the properties of bodies. They also precede bodies, forming the conditions in which bodies develop and live. There is, as Celia Roberts (2000) puts it, a co-construction of the biological and the social.

Starkly opposed to the body as machine is the idea of the body as a canvas on which culture

paints images of womanhood and manhood. Second-wave feminism was very much concerned with the way women's bodies were represented and moulded. One of the very first Women's Liberation demonstrations was against the Miss America beauty pageant at Atlantic City in 1968. (Contrary to an almost universal media myth, no bras were burned at this demonstration. Rather, bras and other constricting underclothes were thrown into a 'Freedom Trash Can'.)

Research on gender imagery is abundant. Historical studies such as Lois Banner's *American Beauty* (1983) trace the changing systems of signs through which women's bodies are defined as elegant, beautiful and desirable, or unfashionable and ugly. The imagery of men's bodies has come in for more recent scrutiny. A nice example is Dorinne Kondo's (1999) study which looks at the advertising campaign through which a Japanese manufacturer of expensive suits created an appeal to a supposed Japanese aesthetic, embodied by the elite of salarymen who stalked around in their particular brand.

Recent cultural studies of the body often focus on language or discourse, under the influence of the French historian Michel Foucault. In a number of celebrated studies, most completely in *Discipline and Punish* (1977), Foucault showed how modern systems of knowledge had come to sort people into categories, and how these categories were interwoven with techniques of social discipline that policed their bodies. The categories and the discipline were applied by professions such as medicine, psychology and criminology, in an amalgam that Foucault called power/knowledge (it rhymes in French: *pouvoir-savoir*).

Foucault, notoriously, failed to theorize gender, though most of his stuff is actually about men in masculinized institutions. However, his approach was taken up by many post-structuralist feminists (e.g. Fraser 1989), and is readily turned into a theory of gender by treating gendered bodies as the products of disciplinary practices. Bodies are 'docile' and biology bends to the hurricane of social discipline.

Field research shows how such disciplining is done. The Los Angeles body-building gyms studied by the ethnographer Alan Klein (1993) reveal a whole subculture of men subjected to a fierce regime of exercise, diet and drugs. Over years of subjection to this regime their bodies are sculpted into the ideal masculine forms desired in body-building competitions. This is an extreme case, but more moderate disciplining of bodies is very widespread. It is undertaken by such powerful institutions as sport, education and medicine. Nancy Theberge (1991) shows how the different exercise regimes for men and women, the disciplinary practices that both teach and constitute sports, are designed to produce gendered bodies. Michael Messner (2007), meanwhile, has shown, through a long research programme, how pervasive are gender stereotyping and gender inequality in the US sports world.

And if social discipline cannot produce gendered bodies, the knife can. The silicon breast implant scandal made public the scale on which cosmetic surgery has been done in the United States, where big breasts are thought sexy. This whole industry, one might think, flies in the face of the ideology of natural difference. Research on cosmetic surgeons and their clients by Diana Dull and Candace West (1991) shows a startling solution. Cosmetic surgery has become considered 'natural' for a woman, but not for a man. The exception is penile surgery, where penis enlargement is now a considerable business. However, a growing number of men in the

global North are also undertaking procedures like rhinoplasty (nose jobs), breast reduction, eyelid surgery and liposuction.

Body-as-canvas approaches, though they have been wonderfully productive, also run into difficulty. The approach emphasizes the signifier to the point where the signified practically vanishes. With gender, the difficulty is crucial. What makes a symbolic structure a gender structure, rather than some other kind, is the fact that its signs refer directly or indirectly to the way humans reproduce.

Post-structuralist research often exaggerates the docility of bodies. Bodies may participate in disciplinary regimes not because they are docile, but because they are active. They seek pleasure, experience, transformation. Some startling examples of this can be found in sadomasochist sexual subcultures. People submit to corsets, chains, piercing, branding, rope bondage and a whole spectrum of painfully restrictive clothes in rubber and leather – voluntarily, indeed with delight (Steele 1996). The same is surely true, in milder forms, of the whole system of fashion. Nobody compels young women to wear shoes with stiletto heels. They hurt after a few minutes, and they wreak havoc after a few years, but they are also enjoyable.

Some feminist philosophers, such as Elizabeth Grosz, have argued there is no consistent distinction between body and mind, and that our embodiment itself is adequate to explain our subjectivity. In her book *Volatile Bodies: Towards a Corporeal Feminism* (1994), Grosz uses ideas from a range of Western European philosophers, such as Jacques Lacan, Sigmund Freud, Maurice Merleau-Ponty, Friedrich Nietzsche and Luce Irigaray, to critique the mind/body dualism. She argues that bodies and minds are two aspects of the one element making up ourselves and the world, and uses active language to describe gendered embodiment as dynamically shifting over time. On the question of sexual difference, Grosz maintains that there is an irreducible distinction between sexes.

Bodies are also recalcitrant and difficult. They are vulnerable to epidemic disease. The HIV/AIDS pandemic, one of the great challenges of our time, is interwoven with embodied gender relations and gender practices on a global scale, from domestic violence to forms of sexuality (Schneider and Stoller 1995). As Wendy Harcourt shows in her powerful book *Body Politics in Development* (2009), a whole series of issues in global economic development involve troubles about bodies – in childbirth and childcare, violence, sexualization, technology and more.

Studies of men in industries such as construction and steelmaking show another side of embodiment. The masculinity of industrial labour in these settings consists in its heaviness, risk and difficulty, where men put themselves ‘in harm's way’, as a vivid ethnographic study of the US construction industry puts it (Paap 2006). In such labour, bodies are consumed: worn down, injured, sometimes killed. As Mike Donaldson (1991) remarks, ‘The very destruction of the physical site of masculinity, the body, can be a method of attaining, demonstrating and perpetuating the socially masculine.’

So bodies cannot be understood as just the objects of social process, whether symbolic or disciplinary. They are active participants in social process. They participate through their

capacities, development and needs, through their recalcitrance, and through the directions set by their pleasures and skills. Bodies share in social agency, in generating and shaping courses of social conduct. Yet all the difficulties of biological determinism, outlined above, remain. Can we solve these problems by holding both a machine and a canvas image of the gendered body at the same time?

In the 1970s a number of feminist theorists did exactly this, proposing a sharp distinction between 'sex' and 'gender'. Sex was the biological fact, the difference between the male and the female human animal. Gender was the social fact, the difference between masculine and feminine roles, or men's and women's personalities.

To many at the time, this two-realms model was a conceptual breakthrough, showing why biology could not be used to justify women's subordination. The constraints of biological difference were confined to the realm of biology itself. A broad realm of social life ('culture', 'roles', etc.) remained, a realm of freedom, where individuals or societies could choose the gender patterns they wanted. Eleanor Maccoby and Carol Jacklin, the authors of a vast and influential survey of *The Psychology of Sex Differences*, concluded:

We suggest that societies have the option of minimizing, rather than maximizing, sex differences through their socialization practices. A society could, for example, devote its energies more toward moderating male aggression than toward preparing women to submit to male aggression, or toward encouraging rather than discouraging male nurturance activities. (1975: 374)

The concept of 'androgyny' put forward by Sandra Bem (1974) and other psychologists at this time was a popular attempt to define an alternative gender pattern. Androgyny meant a mixture of masculine and feminine characteristics, which an individual or a society could choose.

At the high tide of North American liberal feminism in the 1970s, the two-realms model supported an optimistic view of change. Oppressive gender arrangements, being the products of past choices, could be abolished by fresh choices. Whole reform agendas were constructed around this principle. Among them were media reforms (to change sex role models), educational reforms (to change the expectations transmitted to girls and boys) and new forms of psychotherapy (to help individuals make the change to new roles).

A notable example of educational reform is the pioneering Australian Schools Commission's report *Girls, School and Society* (1975). This described the ways girls were held back by restrictive social stereotypes, and proposed action to break down educational segregation and to widen girls' job choices. From this report flowed many projects in Australian schools encouraging girls to work in areas such as mathematics, science and technology.

However, the two-realms model soon ran into trouble, as Rosemary Pringle (1992) shows in a careful critique. The idea of gender as culturally chosen difference ('sex roles') was unable to explain why one side of that difference, the masculine, was more highly valued than the other. The separation of gender from bodies ran counter to developments in feminism which were placing stronger emphasis on bodies.

If the two realms cannot be held strictly apart, perhaps they can be added together? A commonsense compromise would suggest that gender differences arise from *both* biology and social norms. This additive conception underlies most discussions of gender in social psychology, where the term 'sex role' is still widely used. This very phrase adds together a biological and a dramaturgical term.

But there are difficulties in the additive conception too. The two levels of analysis are not easily comparable. In such discussions, it is almost always assumed that biology's reality is more real than sociology's, its explanations more powerful, and its categories more fixed. The passage from Maccoby and Jacklin quoted above continues, saying: 'A variety of social institutions are viable within the framework set by biology.' Maccoby and Jacklin argue for social choice, and want change, but the priority in their analysis is clear. Biology determines; only within its 'framework' may humans choose their gender arrangements. Sex role theory and sex difference research constantly collapse back into biological dichotomy.

A further difficulty is that the patterns of difference at the two levels need not match. As we have seen already, human bodies are dimorphic only in limited ways. On the other side, human behaviour is hardly dimorphic at all, even in areas closely related to sexual reproduction. For instance, while few men do childcare with infants, it is also true that, at any given time, most women are not doing this work either.

In current social life there is a whole spectrum of gender variations. In *Breaking the Bowls* (2005), the US sociologist Judith Lorber observes that for any individual, gender is composed of: sex category; gender identity; gendered marital and procreative status; gendered sexual orientation; gendered personality; gender processes (in everyday interaction); gender beliefs; and gender display. Since there is variety in most of these elements, the number of available gender positions rises into the hundreds, perhaps into the thousands. So much for 'dimorphism'!

It is impossible to sustain a two-realms model of gender difference, any more than we can sustain the machine or canvas models. It is time to look more closely at the evidence about difference itself.

Facts about difference: 'sex similarity' research

In pop psychology, bodily differences and social effects are linked through the idea of *character dichotomy*. Women are supposed to have one set of traits, men another. Women are supposed to be nurturant, suggestible, talkative, emotional, intuitive and sexually loyal; men are supposed to be aggressive, tough-minded, taciturn, rational, analytic and promiscuous. These ideas have been strong in European-origin cultures since the nineteenth century, when the belief that women had weaker intellects and less capacity for judgement than men was used to justify their exclusion from universities and from the vote.

The belief in character dichotomy was one of the very first issues about gender to be addressed in empirical research. Starting in the 1890s, generations of psychologists have

measured various traits with tests or scales, and compared the results for women with those for men. This body of research, long known as ‘sex difference’ (sometimes ‘gender difference’) research, is huge; this is one of the most-researched topics in psychology. There is also a large parallel literature in sociology and political science, looking at group differences in attitudes and opinions, voting, violence, and so forth.

The beginning of this research is described in a fascinating historical study by Rosalind Rosenberg, *Beyond Separate Spheres* (1982). The first generation of psychological researchers found, contrary to mainstream nineteenth-century belief, that the mental capacities of men and women were more or less equal. It is an interesting fact that this finding of ‘no difference’ was rapidly accepted by men as well as women in the mental-testing field. Indeed, as they developed standardized tests of general ability or intelligence (the so-called ‘IQ tests’) during the first half of the twentieth century, psychologists incorporated the ‘no difference’ finding as a given. They chose and scored test items in such a way that males and females would have equal average scores. Later attempts to find gender differences in this field have come to nothing (Halpern and LaMay 2000). It is now widely accepted that in general intelligence, there are no significant gender differences.

An even more interesting fact is that this is the usual finding in the gender difference research on other variables too. In table after table of Maccoby and Jacklin's book, the commonest entry in the column for the finding about difference is ‘none’. Study after study, on trait after trait, comparing women's results with men's, or girls' with boys', finds no significant difference. In summarizing their findings, the first thing Maccoby and Jacklin (1975: 349) did was list a series of ‘unfounded beliefs about sex differences’. On the evidence they compiled, it is *not* true that girls are more social than boys, that girls are more suggestible than boys, that girls have lower self-esteem, that girls are better at rote learning and boys at higher-level cognitive processing, that boys are more analytic, that girls are more affected by heredity and boys by environment, that girls lack achievement motivation, or that girls are auditory while boys are visual. All these beliefs turn out to be myths.

Maccoby and Jacklin were not alone. Other reviewers too noted that the main research finding is a massive psychological *similarity* between women and men in the populations studied by psychologists. If it were not for the cultural bias of both writers and readers, we might long ago have been talking about this as ‘sex *similarity*’ research (Connell 1987: 170).

It is therefore intensely interesting to find that this conclusion is widely disbelieved. The acceptance of gender similarity in the field of intelligence testing turns out to have been exceptional. Most people still believe in character dichotomy. Pop psychology is utterly committed to this idea. In the academic world, generations of researchers, in the teeth of the evidence their own disciplines have produced, have gone on relentlessly searching for, and writing about, psychological gender differences.

The gap between the main pattern *actually* found, and the widespread belief about what *should* be found, is so great that Cynthia Epstein entitled her admirable book about dichotomous thinking and gender reality *Deceptive Distinctions* (1988). Two decades later, when she was President of the American Sociological Association, Epstein (2007) was still

having to argue against the conventional 'master narrative' that holds 'that men and women are naturally different and have different intelligences, physical abilities, and emotional traits'.

Why the huge reluctance to accept the evidence of similarity? A large part of the explanation lies in the cultural background. Dichotomous gender symbolism is very strong in European culture, so it is not surprising that when researchers look at sex and gender, what they 'see' is difference. Within our usual research design, gender similarity is not a positive state; it is merely the absence of proven difference (literally, the 'null hypothesis'). Epstein gives an example of journal editors not liking to publish null results. So the true evidence for gender similarity may be *even stronger* than the published literature reveals.

But are the facts as solid as they seem? Conventional psychological tests, it is sometimes said, are too superficial to detect the underlying patterns of gender. The real character differences between women and men may be lodged at a deeper level in personality – say, in the unconscious. (Jungian psychology makes a dichotomy of the 'deep masculine' and the 'deep feminine'.) This could be true. Certainly most quantitative tests in psychology measure only the immediately apparent aspects of behaviour, often through self-report. But if the 'deep' differences don't show up at the level of everyday life, and keep on failing to show up across a wide range of behaviours – which is what the quantitative research reveals – then one wonders how important such deep differences really can be.

A second issue is that the finding of 'no difference' is not uniform. Maccoby and Jacklin also pointed to a small number of traits where gender differences *did* exist, according to the bulk of their evidence: verbal ability, visual-spatial ability, mathematical ability and aggressiveness. It was these findings, not the larger 'no difference' finding, which went into the textbooks, and have been emphasized by most subsequent writers.

A third issue concerns research method. Maccoby and Jacklin had a large amount of data, but most of it came from hundreds of small studies with ill-defined samples. It may be that the number of 'no difference' findings reflects the methodological weakness of the individual studies. If a way could be found to strengthen the method by combining the results of many studies, the picture might change.

Exactly this became possible when a new statistical procedure, known as 'meta-analysis', was introduced to gender difference research in the 1980s. The procedure relies on finding a large number of separate studies of the same issue: for instance, many studies attempting to measure gender differences in aggression, or intelligence, or self-esteem. In meta-analysis each study (rather than each person) is taken as one data point, and the task is to make a statistical analysis of the whole set of studies.

Obviously before this can be done the findings have to be expressed on a common scale. Unless all the studies have used exactly the same measurement procedures, which in practice is rarely the case, this is a problem. The ingenious solution is to define a common scale based on the variability of individual scores in the original studies. For each study, the difference between the average scores of women and men (on whatever test is being used) is obtained, and this is rewritten as a fraction of the overall variation in people's scores found in that study on that same test. (Technically, the difference between means is divided by the mean within-

group standard deviation.) The standardized gender difference, known as ' d ', found for each individual study is the measurement taken forward into the meta-analysis. By an unfortunate convention, a positive d means that men as a group score higher; a negative d means that women as a group score higher.

The first impact of meta-analysis was to revive confidence in the existence and importance of gender differences generally. This can be seen in Alice Eagly's *Sex Differences in Social Behavior* (1987). Even when most studies in a group individually show non-significant differences, meta-analysis may find an effect size significantly different from zero in the group as a whole. A few examples from the many effect sizes reported are: +.21 across 216 studies of self-esteem (Kling et al. 1999), -.28 across 160 studies of 'care orientation' in moral choice (Jaffee and Hyde 2000), +.48 across 83 studies of aggression (Hyde 1984), zero across 22 studies of 'meaning orientation' in learning styles (Severiens and ten Dam 1998), and between .01 and .17 in large surveys of the frequency of sexual intercourse (Petersen and Hyde 2011).

The question then arises, what do these effect sizes mean? An effect may be significantly different from zero, which means it is not a result of pure chance, but may still be so small that it does not tell us much about the world. Here meta-analysis has its limits. By convention, an effect size of .20 is called 'small', .50 is called 'medium', and .80 is called 'large'. But there is debate about how to interpret this convention. Eagly (1987) argued that even small effects may be practically important, but other meta-analysts are much less convinced.

As meta-analyses built up, so did a renewed scepticism about the size and scope of gender differences. Maccoby and Jacklin in the 1970s considered that 'verbal ability' was one of the traits where a difference (favouring women) was definitely established. But Hyde and McKinley (1997), reviewing meta-analyses of research since then, report effect sizes clustering around zero. Mathematics ability, another claimed area of difference (favouring men), proves to have only very small effect sizes, between -.15 and +.22 for 242 studies, supporting the view that males and females perform similarly (Lindberg et al. 2010).

In 2005 Janet Hyde published a grand meta-survey of meta-analyses, combining the findings of this technique across the whole field of psychology. She found 46 published meta-analyses of gender difference, which analysed over 5,000 research studies, which in turn were based on the testing of about 7 million people. The research covered cognitive variables, communication, social and personality variables, psychological well-being, motor behaviours and assorted other topics. The overall finding is simply stated:

The striking result is that 30% of the effect sizes are in the close-to-zero range, and an additional 48% are in the small range. That is, 78% of gender differences are small or close to zero. (Hyde 2005: 582, 586)

Hyde provocatively titles her paper 'The Gender Similarities Hypothesis'. It is hardly a hypothesis. The idea of a character dichotomy between women and men has been overwhelmingly, decisively, refuted. *The broad psychological similarity of men and women as groups can be regarded, on the volume of evidence supporting it, as one of the best-established generalizations in all the human sciences.*

Hyde also recognizes that there are some traits on which average gender differences do persistently show up. In her review they include physical performances (e.g. in throwing); some aspects of sexuality, but not all; and some aspects of aggression.

What is particularly interesting in the meta-analytic studies is that when clear psychological gender differences do appear, they are likely to be specific and situational rather than generalized. Studies of aggression often show a gender difference – but in physical aggression more than in verbal aggression, and not in all circumstances. Ann Bettencourt and Norman Miller (1996) find an overall d of +.22 in experimental studies of aggression, but report that this effect depends on whether or not there are conditions of provocation. Unprovoked, men have a modest tendency to show higher levels of aggressiveness than women (mean effect size +.33); provoked, men's and women's reactions are similar (mean effect size +.17). Hyde cites a meta-analysis of gender differences in making interruptions in conversation. The effect size varied, according to the type of interruption, the size of the group talking together, and whether they were strangers or friends. She comments: 'Here, again, it is clear that gender differences can be created, erased, or reversed, depending on the context' (Hyde 2005: 589).

Meta-analysis reveals that gender differences in masculinity/femininity, as measured by tests like the 'Bem Sex Role Inventory', change over time. Jean Twenge (1997) showed that men and women (in samples of US undergraduates) became more similar in their responses on these scales over a period of 20 years, from the 1970s to the 1990s. This was not, as many people fear, because men are becoming feminized – both groups' scores on the femininity scales changed little. It was rather because the women increased their scores markedly (and men a little) on masculinity scales over this period.

Meta-analysis has not entirely revolutionized the study of gender difference, since the basic data collection methods remain the same. But it has certainly clarified what this body of research is saying. Confirming earlier conclusions, it tells us that across a wide range of the traits and characteristics measured by psychology, sharp gender differences are rare. Broad similarity between women and men is the main pattern. Meta-analysis adds a clearer recognition that specific and situational sex differences often appear. Very specific skills (e.g. in one science rather than another), specific social circumstances (e.g. provocation), specific times and places (e.g. US colleges in the 1990s) and specific ways of measuring traits all affect the extent of gender differences recorded in the research.

We thus get a picture of psychological gender differences and similarities not as fixed, age-old constants of the species, but as the changing products of the active responses people make to a complex and changing social world. With the aid of meta-analysis, psychology has gradually moved towards the way of understanding gender that has also gradually emerged in sociology.

How far can we generalize this picture? It is often observed that the modern science of psychology is mainly based on the behaviour of white middle-class students in Psychology 101 courses in US universities – not exactly a representative sample of humanity.

Given the impressive evidence of cultural and historical variations in gender arrangements (see [chapters 2](#) and [5](#)), we cannot simply assume that the psychological patterns documented

for the contemporary US hold true across the world. Yet this very point, that gender differences can vary between different circumstances, has been emerging from meta-analytic research. The gender similarity research now includes increasing numbers of large-scale studies with better samples of the population of countries in the global North, and increasing numbers of studies in other parts of the world. The conclusions outlined above are a solid starting point for understanding the psychology of gender.

Social embodiment and the reproductive arena

Now that gender similarity research has decisively refuted the concept of character dichotomy, we must reject all models of gender that assume social gender differences to be caused by bodily differences producing character differences. How, then, can we understand the relation between body and society in gender?

Bodies have agency *and* bodies are socially constructed. Biological and social analysis cannot be cut apart from each other. But neither can be reduced to the other. Within a 'difference' framework, these conclusions sit as paradoxes. We must move towards another framework.

There are many, many differences among the 7,000 million human bodies in the world. There are old and young, sick and well, plump and starving. There are differences of physical ability and disability. There are skins permanently stained with soil and skins softened with expensive creams; hands cracked from washing and hands spotless and manicured. Each body has its trajectory through time; each changes as it grows older. Some bodies encounter accident, traumatic childbirth, violence, starvation, disease or surgery, and have to reorganize themselves to carry on. Some do not survive these encounters.

Yet the tremendous multiplicity of bodies is in no sense a random assortment. Our bodies are interconnected through social practices, the things people do in daily life. Bodies are both *objects of* social practice and *agents in* social practice. The same bodies, at the same time, are both. The practices in which bodies are involved form social structures and personal trajectories, which in turn provide the conditions of new practices in which bodies are addressed and involved. Bodily processes and social structures are linked *through time*. They add up to the historical process in which society is embodied, and bodies are drawn into history.

In this book, this historical process is called *social embodiment*. From the point of view of the body, it could be called 'body-reflexive practice', i.e. human social conduct in which bodies are both agents and objects.

Bodies have a reality that cannot be reduced; they are drawn into history without ceasing to be bodies. They do not turn into signs or positions in discourse (though discourses constantly refer to them). Their materiality continues to matter. We are born, we are mortal. If you prick us, do we not bleed?

Social embodiment involves an individual's conduct, but also involves groups, institutions

and whole complexes of institutions. Consider the body-reflexive practice that goes into making the exemplary masculinity of a sports star – for instance, Steve, a champion in ‘iron man’ surf competitions, whose life is described in *The Men and the Boys* (Connell 2000). Steve's practice includes the training routines worked out by coaches, drawing on the professional expertise of physical education and sports medicine. It includes the practice of the sport itself, which is organized by multi-million-dollar corporations. It includes participating in publicity and getting fees via other corporations (commercial media, advertisers). A major sports star, like other media figures, practically turns into a one-person corporation, employing lawyers, accountants, marketing agents and public relations flacks. There is an elaborate social process here. Yet all of this specialized work is based on, and refers back to, Steve's bodily performances.

Gender is a specific form of social embodiment. The distinctive feature of gender is that it refers to the bodily structures and processes of human reproduction. Gender involves a cluster of human social practices – including childcare, birthing, sexual interaction – which deploy human bodies' capacities to engender, to give birth, to give milk, to give and receive sexual pleasure. We can only begin to understand gender if we understand how closely the social and the bodily processes mesh. We are born in blood and pain, *and* we are born in a social order.

These bodily capacities, and the practices that realize them, constitute an arena, a bodily site where something social happens. Among the things that happen is the creation of the cultural categories ‘women’ and ‘men’ (and any other gender categories that a particular society marks out). This may be called the *reproductive arena* in social life. The beginning of the chapter noted that we are one of the species that reproduce sexually, and this is where that fact becomes central for gender analysis.

The idea of a ‘reproductive arena’ is not the same as the traditional idea of a ‘biological base’, a natural mechanism that produces automatic social effects. Sexual reproduction does not *cause* gender practice, or even provide a template for it. There are many fields where strongly gendered behaviour occurs which has not the slightest logical connection with sexual reproduction (football, shoe design, futures markets, lesbian sex, Handel oratorios, the appointment of bishops ...). We may be one of many species that reproduce sexually, but we are the only one of them that has produced complex, historically changing social structures in which that reproductive capacity is deployed and transformed. Gender, in fact, is one of the most striking things that is unique about our species.

Though it is often forgotten in the excitement of gender politics among adults, the reproductive arena very much concerns children. Not all sex results in pregnancy – in fact the great majority of sexual encounters, even heterosexual ones, don't, and aren't intended to. But the fact that children do arrive this way, and have to be nurtured and taught, and will become the next generation of parents, matters immensely for any society that intends to last beyond next Thursday. The way caring for children is organized is a large part of the domain of gender.

The reproductive arena can be reshaped by social processes. Indeed it constantly is being reshaped in social struggle. For instance, the fertility of a woman's body means something different where contraception is effective and small families are planned from what it means

where women are designated lifelong breeders and nurturers – barefoot, pregnant and in the kitchen, as the saying goes.

The reproductive arena is always the point of reference in gender processes, but it is far from incorporating everything that gender is about. We also need a concept of the *gender domain*, meaning the whole terrain of social life that is socially linked to the reproductive arena. In the gender domain, relations among people and groups are structured by this linkage and can therefore be understood as gender relations.

It follows from this definition that the scope and shape of the gender domain vary from one society to another, and from one period of history to another. They can even be changed by deliberate action. This is attempted in the ‘de-gendering’ reform strategy (Connell 2006; Lorber 2005). The attempt to de-gender early childhood education in rich countries by employing more men as teachers is a current, though so far not very successful, example. (In some poorer countries there are already more men than women in elementary teaching.)

A few short examples of social embodiment in the gender domain may illustrate what is meant. One of the crudest ways of deploying gender is through sexual harassment – an exercise of power, directed to the body of the target. Meredith Newman and her colleagues (2003) report survey evidence from employees of the US government showing 25 per cent of women report having been sexually harassed on the job, compared with 6 per cent of men. There was little variation from one government agency to another. Greetje Timmerman and Cristien Bajema (1999) found that between 17 and 81 per cent of employed women reported experiencing some form of workplace sexual harassment, according to surveys in 11 European countries.

Miriam Glucksmann's (1990) historical study of women in the electrical engineering and food processing industries in inter-war Britain shows that gender segregation was introduced in the new factories on a massive scale. Nothing about the workers' bodies, or about the technology of chocolate biscuit making, required this. But the factory managers sharply segregated women's from men's bodies. The reason was that to have integrated the workplace would have broken down the existing social dependence of women and the gender division of child care and housework in the workers' homes.

The devastating human immunodeficiency virus, which leads to AIDS, has been spread around the world by contacts between human bodies, usually following gendered paths. Purnima Mane and Peter Aggleton (2001), surveying the role of men in this biological process, note that men's practices in different regions are shaped by local gender orders. The risk of transmission is highest where women have least capacity to control sexuality. Men's bodily practices are also shaped by prevailing definitions of manhood, such as the acceptance of risk, and the idea of proving manhood by sexual experimentation. For instance, in sub-Saharan Africa, where there is a mostly heterosexual HIV epidemic, women who experience more marked gender inequality and violence in relationships are at greater risk of HIV. Risks stem from a lack of control over the circumstances of sex. This inequality and violence in relationships is often related to complicity with a power-oriented ideal of hegemonic masculinity (Jewkes and Morrell 2010).

Bodies are transformed in social embodiment. Some broad changes in recent history are familiar: falling numbers of children born, lengthening expectation of life, rising average height and weight (as child nutrition and healthcare improve), and changing patterns of disease (e.g. polio declining, TB declined but now reviving). The transformation of bodies is structured, in part, on gender lines, as the demographic indicators show. In the rich industrial countries, women's average life expectancy is now noticeably longer than men's. The average expectation of life for women in OECD countries is 83 years, for men 77. But in some poorer countries the difference is noticeably less: in Pakistan, the figure for women is 68 years and for men 66; in the Congo, for women 58 and for men 55.

The idea of social embodiment allows us to recognize a paradoxical aspect of gender. Many gender processes involve bodily processes and capacities that are themselves not gender-differentiated, that are in fact common capacities of women and men. For instance, there are almost no gender differences of any consequence in capacities to work in an industrial economy, apart from those created by different training, the treatment of pregnancy as a disability, or the gendered design of equipment. Most production processes involve the co-operation of very large numbers of men and women in an intricate flow of work. Ironically, this shared labour creates the means through which mass media images of gender dichotomy are circulated.

Recognizing social embodiment also allows a new view of the relation between bodies and change in gender. In sociobiology, sex role theory, liberal feminism and popular ideologies of natural difference, bodily difference is understood to be a conservative force. It holds back historical change, limits what social action can accomplish. But we can now see that bodies as agents in social practice are involved in the very construction of the social world, the bringing-into-being of social reality. Bodies' needs, bodily desire and bodily capacities are at work in history. The social world is never simply reproduced; it is always reconstituted by practice.

Gender as a system of relations is created in this historical process, and accordingly can never be fixed, nor exactly reproduced. The strategic question is not 'can gender change?' but 'in what direction is gender changing?' There are different futures towards which contemporary societies might be moved. We will explore this in the later chapters of the book. First, we need to look at the ways gender has been theorized.